Chapter 7

Campbell Biology, AP Edition, 12e

Vesicles fuse with a membrane.

Chapter 7********

Membrane Structure and Function

Figure 7.1

Successful learning relies on communication between brain cells. Here, the vesicles fusing with the plasma membrane of the top cell release molecules (yellow) that bind to membrane proteins (light green) on the surface of the bottom cell, triggering the proteins to change shape. The plasma membrane that surrounds each cell regulates its exchanges with its environment and surrounding cells.

How does the plasma membrane regulate inbound and outbound traffic?

A cell is given. The cell has a plasma membrane. Inside are vesicles. Exocytosis: large molecules are secreted when a vesicle fuses with the plasma membrane. Bulk transport moves large molecules. Endocytosis: large molecules are taken in when the plasma membrane pinches inward, forming a vesicle. Active transport of small molecules requires energy and a transport protein. Passive transport of small molecules doesn't require energy. It may involve transport proteins.

Key Concepts

- **7.1Cellular membranes are fluid mosaics of lipids and proteins****
- **7.2Membrane structure results in selective permeability****
- **7.3Passive transport is diffusion of a substance across a membrane with no energy investment****
- **7.4Active transport uses energy to move solutes against their gradients****

- **7.5Bulk transport across the plasma membrane occurs by exocytosis and endocytosis****

Advanced placement edition

The plasma membrane is a structure conserved across domains **(BIG IDEA 1)** yet highly specialized to species and their internal and external environments **(BIG IDEA 2).** The plasma membrane's emergent properties **(BIG IDEA 3)** regulate the complex interactions of cells **(BIG IDEA 4).**

Study Tip

Make a visual study guide: Draw a plasma membrane (two lines) all the way down a piece of paper (or digitally). Label the cytoplasm and the extracellular fluid. At the top, draw the phospholipid bilayer in detail. As you read the chapter, draw and label membrane proteins that you encounter and diagram the different ways materials can enter or leave a cell.

Part of a cell.

The plasma membrane is a phospholipid bilayer. On one side is the cytoplasm, and on the other is the extracellular fluid. A glycoprotein (carbohydrate plus protein) spans the plasma membrane.

Get Ready for This Chapter

In this chapter, you will learn about the organization, structure, and function of cell membranes, including regulation of transport into and out of the cell. Before beginning this chapter, you should be able to do the following:

- Explain hydrophilic and hydrophobic properties of molecules (see Concept 3.2, section entitled "Hydrophilic and Hydrophobic Substances").
- Describe the structure of a phospholipid (see Concept 5.3, section entitled "Phospholipids").
- Describe the structure of prokaryotic and eukaryotic cells, with a focus on the presence of membranes (see Concept 6.2, Figure 6.5, and Figure 6.8).
- Explain the basic idea of potential energy (see Concept 2.2, section entitled "The Energy Levels of Electrons").

The following questions provide a quick check of your basic knowledge in these areas.

1. Hydrophobic substances like salad oil are

```
     nonpolar molecules that do not have an affinity for
water molecules.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Salad oil is predominantly made up of carbon and hydrogen atoms, which share electrons almost equally, forming nonpolar covalent bonds. Substances that are nonpolar due to their large number of nonpolar bonds do not have an affinity for water and are termed hydrophobic (“water-fearing”). Substances that contain polar bonds are hydrophilic (“water-loving”) because they contain atoms with partial charges due to those polar bonds. For example, a hydrogen atom with a partial positive charge can form a hydrogen bond with a partial negative charge on the oxygen atom of a water molecule.

```
        nonpolar molecules that have an affinity for water
because they contain many hydrogens that can form hydrogen bonds
with water.
```

That's incorrect. Try again.

Incorrect. Review the following material: Salad oil is predominantly made up of carbon and hydrogen atoms, which share electrons almost equally, forming nonpolar covalent bonds. Substances that are nonpolar due to their large number of nonpolar bonds do not have an affinity for water and are termed hydrophobic (“water-fearing”). Substances that contain polar bonds are hydrophilic (“water-loving”) because they contain atoms with partial charges due to those polar bonds. For example, a hydrogen atom with a partial positive charge can form a hydrogen bond with a partial negative charge on the oxygen atom of a water molecule.

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with a partial positive charge can form a hydrogen bond with a partial negative charge on the oxygen atom of a water molecule.

```
polar molecules that have an affinity for water molecules because they contain many hydrogens that can form hydrogen bonds with water.
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That's incorrect. Try again.

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2. How do membrane phospholipids interact with water?

```
Phospholipids do not interact with water because they are lipids, and thus are completely hydrophobic.
```

That's incorrect. Try again.

Incorrect. Review the following material: A phospholipid is similar to a fat molecule but has only two fatty acids attached to glycerol rather than three. The third hydroxyl group of glycerol is joined to a phosphate group, which has a negative electrical charge in the cell. The fatty acids, referred to as the “tails” of the phospholipid, are hydrocarbons that are hydrophobic and therefore do not interact with water. The phosphate group and its attachments form a hydrophilic “head” that has an affinity for water.

```
They have hydrophilic tails that face outward and are exposed to water and hydrophobic heads that face the center of the membrane and are shielded from water.
```

That's incorrect. Try again.

Incorrect. Review the following material: A phospholipid is similar to a fat molecule but has only two fatty acids attached to glycerol rather than three. The third hydroxyl group of glycerol is joined to a phosphate group, which has a negative electrical charge in the cell. The fatty acids, referred to as the “tails” of the phospholipid, are hydrocarbons that are hydrophobic and therefore do not interact with water. The phosphate group and its attachments form a hydrophilic “head” that has an affinity for water.

```
The polar heads have no affinity for water and the nonpolar tails have an affinity for water.
```

That's incorrect. Try again.

Incorrect. Review the following material: A phospholipid is similar to a fat molecule but has only two fatty acids attached to glycerol rather than three. The third hydroxyl group of glycerol is joined to a phosphate group, which has a negative electrical charge in the cell. The fatty acids, referred to as the “tails” of the phospholipid, are hydrocarbons that are hydrophobic and therefore do not interact with water. The phosphate group and its attachments form a hydrophilic “head” that has an affinity for water.

```
The polar heads interact with water; the nonpolar tails do not.
```

That's correct.

Correct! If you had trouble with this question, review the following material: A phospholipid is similar to a fat molecule but has only two fatty acids attached to glycerol rather than three. The third hydroxyl group of glycerol is joined to a phosphate group, which has a negative electrical charge in the cell. The fatty acids, referred to as the “tails” of the phospholipid, are hydrocarbons that are hydrophobic and therefore do not interact with water. The phosphate group and its attachments form a hydrophilic “head” that has an affinity for water.

3. Which of the following is a major difference between prokaryotic and eukaryotic cells?

```
Prokaryotic cells have no DNA; eukaryotic cells have DNA.
```

That's incorrect. Try again.

Incorrect. Review the following material: All cells are enclosed by a plasma membrane, a selective barrier that forms a boundary between the cell and the external environment. The existence of cells depends on the properties of phospholipids that make up the plasma membrane. Eukaryotic cells have extensive, elaborately arranged internal membranes as well that divide the cell into compartments called organelles, but most prokaryotic cells do not have internal membranes. All cells have DNA that stores the genetic information of the cell.

```
        Eukaryotic cells have a plasma membrane, whereas
prokaryotic cells do not.
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That's incorrect. Try again.

Incorrect. Review the following material: All cells are enclosed by a plasma membrane, a selective barrier that forms a boundary between the cell and the external environment. The existence of cells depends on the properties of phospholipids that make up the plasma membrane. Eukaryotic cells have extensive, elaborately arranged internal membranes as well that divide the cell into compartments called organelles, but most prokaryotic cells do not have internal membranes. All cells have DNA that stores the genetic information of the cell.

```
        Most prokaryotic cells have no internal membranes;
eukaryotic cells do.
```

That's correct.

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```
        Prokaryotic cells have no phospholipids; eukaryotic
cells do.
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That's incorrect. Try again.

Incorrect. Review the following material: All cells are enclosed by a plasma membrane, a selective barrier that forms a boundary between the cell and the external environment. The existence of cells depends on the properties of phospholipids that make up the plasma membrane. Eukaryotic cells have extensive, elaborately arranged internal membranes as well that divide the cell into compartments called organelles, but most prokaryotic cells do not have internal membranes. All cells have DNA that stores the genetic information of the cell.

4. Which of the following statements is true regarding potential energy?

```
Matter has a natural tendency to acquire more potential energy until a maximum is reached.
```

That's incorrect. Try again.

Incorrect. Review the following material: Energy is defined as the capacity to cause change—for instance, by doing work. Potential energy is the energy that matter possesses because of its location or structure. Matter has a natural tendency to move from higher states of potential energy toward the lowest possible state of potential energy, such as water running downhill from a dam. As water runs downhill, the energy released can be used to do work.

```
Potential energy is the energy possessed by matter due to its location or structure.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Energy is defined as the capacity to cause change—for instance, by doing work. Potential energy is the energy that matter possesses because of its location or structure. Matter has a natural tendency to move from higher states of potential energy toward the lowest possible state of potential energy, such as water running downhill from a dam. As water runs downhill, the energy released can be used to do work.

```
Potential energy is the energy matter could have if it were in a different location or structure.
```

That's incorrect. Try again.

Incorrect. Review the following material: Energy is defined as the capacity to cause change—for instance, by doing work. Potential energy is the energy that matter possesses because of its location or structure. Matter has a natural tendency to move from higher states of potential energy toward the lowest possible state of potential energy, such as water running downhill from a dam. As water runs downhill, the energy released can be used to do work.

```
Water acquires potential energy as it runs downhill.
```

That's incorrect. Try again.

Incorrect. Review the following material: Energy is defined as the capacity to cause change—for instance, by doing work. Potential energy is the energy that matter possesses because of its location or structure. Matter has a natural tendency to move from higher states of potential energy toward the lowest possible state of potential energy, such as water running downhill from a dam. As water runs downhill, the energy released can be used to do work.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 7.1Explain the fluid mosaic model, describing the components of the membrane.
- 7.2Discuss how membrane structure results in selective permeability.
- 7.3Use examples to demonstrate the processes of diffusion, osmosis, and facilitated diffusion.
- 7.4Describe the process of active transport.

Concept 7.1: Cellular membranes are fluid mosaics of lipids and proteins

Lipids and proteins are the staple ingredients of membranes, although carbohydrates are also important. The most abundant lipids in most membranes are phospholipids. Their ability to form membranes is inherent in their molecular structure. A phospholipid is an **amphipathic** molecule, meaning it has both a hydrophilic (“water-loving”) region and a hydrophobic (“water-fearing”) region (see Figure 5.11). A phospholipid bilayer can exist as a stable boundary between two aqueous compartments because the

molecular arrangement shelters the hydrophobic tails of the phospholipids from water while exposing the hydrophilic heads to water **(Figure 7.2)**.

Figure 7.2

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Phospholipid bilayer (cross section).
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Cross section of the phospholipid bilayer.

Phospholipid bilayer with hydrophilic heads drawn as circles and two squiggly hydrophobic tails attached to each head. Tails point towards each other so that only the heads contact water. Cutaway diagram zooms in on the structure of two molecules in the phospholipid bilayer, where hydrophilic heads interact with water molecules on both sides of the membrane, but the hydrophobic tails are tucked away from water.

VISUAL SKILLS *Consulting*Figure 5.11*, circle and label the hydrophilic and hydrophobic portions of the enlarged phospholipids on the right. Explain what each portion contacts when the phospholipids are in the plasma membrane.*

*For suggested answer, see*Appendix A.**

Like membrane lipids, most membrane proteins are amphipathic. Such proteins can reside in the phospholipid bilayer with their hydrophilic regions protruding. This molecular orientation maximizes contact of hydrophilic regions of a protein with water in the cytosol and extracellular fluid, while providing their hydrophobic parts with a nonaqueous environment. **Figure 7.3** shows the currently accepted model of the arrangement of molecules in the plasma membrane. In this **fluid mosaic model**, the membrane is a mosaic of protein molecules bobbing in a fluid bilayer of phospholipids.

Figure 7.3

Current model of an animal cell's plasma membrane (cutaway view). Lipids are colored gray and gold, proteins purple, and carbohydrates green.

Structures associated with the cell membrane in the fluid mosaic model.

Included are membrane proteins, cholesterol, glycoproteins, carbohydrates, glycolipids, fibers of the E C M and microfilaments of the cytoskeleton. The membrane is represented as a phospholipid bilayer, with phospholipid molecules represented as circles with two squiggly tails attached. The tails point towards each other while the head contact both the cytoplasmic side of the cell and the extra cellular matrix (E C M). Embedded in the bilayer are cholesterol molecules (within the bilayer), integral proteins that span the bilayer and peripheral proteins that contact only the cytoplasmic side of the bilayer. Some proteins are

attached to fibers of the E C M or to the cytoskeleton. Inside the cell is a tangle of microfilaments of the cytoskeleton. Outside the cell is a mass of fibers of the E C M. Also, on the extracellular (outer) side of the membrane are glycoproteins, carbohydrates and glycolipids.

Animation

Structure of the Plasma Membrane

image

Play

image

Play

The proteins are not randomly distributed in the membrane, however. Groups of proteins are often associated in long-lasting, specialized patches, where they carry out common functions. Researchers have found specific lipids in these patches as well and have proposed naming them *lipid rafts*; however, the debate continues about whether such structures exist in living cells or are an artifact of biochemical techniques. In some regions, the membrane may be much more tightly packed with proteins than shown in Figure 7.3. Like all models, the fluid mosaic model is continually being refined as new research reveals more about membrane structure.

The Fluidity of Membranes

Membranes are not static sheets of molecules locked rigidly in place. A membrane is held together mainly by **hydrophobic interactions**, which are much weaker than covalent bonds (see Figure 5.18). Most of the lipids and some proteins can shift about sideways—that is, in the plane of the membrane, like partygoers elbowing their way through a crowded room. Very rarely, also, a lipid may flip-flop across the membrane, switching from one phospholipid layer to the other.

The sideways movement of phospholipids within the membrane is rapid. Adjacent phospholipids switch positions about 107 times per second, which means that a phospholipid can travel about 2 μm—the length of a typical bacterial cell—in 1 second. Proteins are much larger than lipids and move more slowly, when they do move. Many membrane proteins seem to be held immobile by their attachment to the cytoskeleton or to the extracellular matrix (see Figure 7.3). Some membrane proteins seem to move in a highly directed manner, perhaps driven along cytoskeletal fibers by motor proteins. And other proteins simply drift in the membrane, as shown the classic experiment described in **Figure 7.4**.

Figure 7.4********

Inquiry: Do membrane proteins move?

Experiment

Larry Frye and Michael Edidin, at Johns Hopkins University, labeled the plasma membrane proteins of a mouse cell and a human cell with two different markers and fused the cells. Using a microscope, they observed the markers on the hybrid cell.

Results

Mouse and human cell, with different membrane proteins. Hybrid cell shows the two cells fused. After one hour, there is one cell with a mixture of proteins from both cells.

Conclusion

The mixing of the mouse and human membrane proteins indicates that at least some membrane proteins move sideways within the plane of the plasma membrane.

Data from L. D. Frye and M. Edidin, The rapid intermixing of cell surface antigens after formation of mouse-human heterokaryons, *Journal of Cell Science* 7:319 (1970).

WHAT IF? *Suppose the proteins did not mix in the hybrid cell, even many hours after fusion. Could you conclude that proteins don't move within the membrane? What other explanation could there be?*

*For suggested answer, see*Appendix A.**

A membrane remains fluid as temperature decreases until the phospholipids settle into a closely packed arrangement and the membrane solidifies, much as bacon grease forms lard when it cools. The temperature at which a membrane solidifies depends on the types of lipids it is made of. As the temperature decreases, the membrane remains fluid to a lower temperature if it is rich in phospholipids with unsaturated hydrocarbon tails (see Figures 5.10 and 5.11). Because of kinks in the tails where double bonds are located, unsaturated hydrocarbon tails cannot pack together as closely as saturated hydrocarbon tails, making the membrane more fluid **(Figure 7.5a)**.

Figure 7.5

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Factors that affect membrane fluidity.
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Factors that affect membrane fluidity.

Part a. Compares unsaturated versus saturated hydrocarbon tails.

- Fluid membrane: Unsaturated hydrocarbon tails (kinked) prevent packing, enhancing membrane fluidity. Tails in phospholipid molecules are drawn askew (kinked), creating spaces in the bilayer.
- Viscous membrane: Saturated hydrocarbon tails pack together, increasing membrane viscosity. Phospholipid molecule tails are drawn as unkinked, without spaces in the bilayer.

Part b. Cholesterol within the animal cell membrane. Cholesterol reduces membrane fluidity at moderate temperatures by reducing phospholipid movement, but at low temperatures, it hinders solidification by disrupting the regular packing of phospholipids. Cholesterol molecules are drawn as chains of four fused rings interspersed with phospholipids in the bilayer.

The steroid cholesterol, which is wedged between phospholipid molecules in the plasma membranes of animal cells, has different effects on membrane fluidity at different temperatures **(Figure 7.5b)**. At relatively high temperatures—at 37°C, the body temperature of humans, for example—cholesterol makes the membrane less fluid by restraining phospholipid movement. However, because cholesterol also hinders the close packing of phospholipids, it lowers the temperature required for the membrane to solidify. Thus, cholesterol can be thought of as a “fluidity buffer” for the membrane, resisting changes in membrane fluidity that can be caused by changes in temperature. Compared to animals, plants have very low levels of cholesterol; rather, related steroid lipids buffer membrane fluidity in plant cells.

Membranes must be fluid to work properly; the fluidity of a membrane affects both its permeability and the ability of membrane proteins to move to where their function is needed. Usually, membranes are about as fluid as olive oil. When a membrane solidifies, its permeability changes, and enzymatic proteins in the membrane may become inactive if their activity requires movement within the membrane. However, membranes that are too fluid cannot support protein function either. Therefore, extreme environments (for example, those with extreme temperatures) pose a challenge for life, resulting in evolutionary adaptations that include differences in membrane lipid composition.

Evolution of Differences in Membrane Lipid Composition

EVOLUTION Variations in the cell membrane lipid compositions of many species appear to be evolutionary adaptations that maintain the appropriate membrane fluidity under specific environmental conditions. For instance, fishes that live in extreme cold have membranes with a high proportion of

unsaturated hydrocarbon tails, enabling their membranes to remain fluid in spite of the low temperature (see Figure 7.5a). At the other extreme, some bacteria and archaea thrive at temperatures greater than 90°C (194°F) in thermal hot springs and geysers. Their membranes include unusual lipids that may prevent excessive fluidity at such high temperatures.

The ability to change the lipid composition of cell membranes in response to changing temperatures has evolved in organisms that live where temperatures vary. In many plants that tolerate extreme cold, such as winter wheat, the percentage of unsaturated phospholipids increases in autumn, an adjustment that keeps the membranes from solidifying during winter. Some bacteria and archaea also exhibit different proportions of unsaturated phospholipids in their cell membranes, depending on the temperature at which they are growing. Overall, natural selection has apparently favored organisms whose mix of membrane lipids ensures an appropriate level of membrane fluidity for their environment.

Membrane Proteins and Their Functions

Now we come to the *mosaic* aspect of the fluid mosaic model. Somewhat like a tile mosaic (shown here), a membrane is a collage of different proteins, often clustered together in groups, embedded in the fluid matrix of the lipid bilayer (see Figure 7.3). More than 50 kinds of proteins have been found in the plasma membrane of red blood cells, for example. Phospholipids form the main fabric of the membrane, but proteins determine most of the membrane's functions. Different types of cells contain different sets of membrane proteins, and the various membranes within a cell each have a unique collection of proteins.

A tile mosaic.

Tile mosaic.

Notice in Figure 7.3 that there are two major populations of membrane proteins: integral proteins and peripheral proteins. **Integral proteins** penetrate the hydrophobic interior of the lipid bilayer. The majority are **transmembrane proteins**, which span the membrane; other integral proteins extend only partway into the hydrophobic interior. The hydrophobic regions of an integral protein consist of one or more stretches of nonpolar amino acids (see Figure 5.14), typically 20–30 amino acids in length, usually coiled into α helices **(Figure 7.6)**. The hydrophilic parts of the molecule are exposed to the aqueous solutions on either side of the membrane. Some proteins also have one or more hydrophilic channels that allow passage through the membrane of hydrophilic substances (even of water itself). **Peripheral proteins** are not embedded in the lipid bilayer at all; they are loosely bound to the surface of the membrane, often to exposed parts of integral proteins (see Figure 7.3).

Figure 7.6

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The structure of a transmembrane protein.
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The structure of a transmembrane protein.

A protein is embedded between the extracellular side and cytoplasmic side of the membrane. On the cytoplasmic side is the C-terminus of the alpha helices, and on the extracellular side is the n-terminus.

Bacteriorhodopsin (a bacterial transport protein) has a distinct orientation in the membrane, with its N-terminus outside the cell and its C-terminus inside. This ribbon model highlights the secondary structure of the hydrophobic parts, including seven transmembrane α helices, which lie mostly within the hydrophobic interior of the membrane. The nonhelical hydrophilic segments are in contact with the aqueous solutions on the extracellular and cytoplasmic sides of the membrane.

On the cytoplasmic side of the plasma membrane, some membrane proteins are held in place by attachment to the cytoskeleton. And on the extracellular side, certain membrane proteins may attach to materials outside the cell. For example, in animal cells, membrane proteins may be attached to fibers of the extracellular matrix (see Figure 6.28; *integrins* are one type of integral, transmembrane protein). These attachments combine to give animal cells a stronger framework than the plasma membrane alone could provide.

Figure 7.7 illustrates six major functions performed by proteins of the plasma membrane. A single cell may have different membrane proteins that carry out various functions, and one protein may itself carry out multiple functions. Thus, the membrane is a functional mosaic as well as a structural one.

Figure 7.7

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Some functions of membrane proteins.
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Functions of membrane proteins.

In many cases, a single protein performs multiple tasks.

Parts are as follows.

Part a, transport. Image shows two proteins embedded in phospholipid bilayer. Drawing on left: A protein that spans the membrane may provide a hydrophilic channel across the membrane that is selective for a particular solute. Through one a solute crosses from high to low concentration. Drawing on right: other transport proteins shuttle a substance from one side to the other by changing shape (see figure 7.14b). Some of these proteins hydrolyze A T P as an energy source to actively pump substances across the membrane. This protein uses A T P to pump a different solute from low to high concentration.

Part b, enzymatic activity. A protein built into the membrane may be an enzyme with its active site exposed to substances in the adjacent solution. In some cases, several enzymes in a membrane are organized as a team that carries out sequential steps of a metabolic pathway. Drawing shows two adjacent enzymes in the membrane carrying out sequential steps in a metabolic pathway, with arrows indicating that the products of one pathway move on to the next enzyme.

Part c, signal transduction. A membrane protein (receptor) may have a binding site with a specific shape that fits the shape of a chemical messenger, such as hormone. The external messenger (signaling molecules) may cause the protein to change shape, allowing it to relay the message to the inside of the cell, usually by binding to a cytoplasmic protein (see figure 11.6).

Part d, cell-cell recognition. Some glycoproteins serve as identification tags that are specifically recognized by membrane proteins of other cells. This type of cell-cell binding is usually short-lived compared to intercellular joining. Image: A membrane protein in one membrane shows a shape that allows it to interact with a glycoprotein in an adjacent cell membrane.

Part e, intercellular joining. Membrane proteins of adjacent cells may hook together in various kinds of junctions, such as gap junctions or tight junctions (see figure 6.30). This kind of binding is more long-lasting than in cell-cell recognition.

Part f, attachment of the cytoskeleton and extracellular matrix (E C M). Microfilaments or other elements of the cytoskeleton may be noncovalently bound to membrane proteins, a function that helps maintain cell shape and stabilizes the location of certain membrane proteins. Proteins that can bind to E C M molecules coordinate extracellular and intracellular changes (see figure 6.28). Image: Membrane proteins bound to microfilaments of the cytoskeleton or two E C M molecules.

VISUAL SKILLS *Some transmembrane proteins can bind to a particular ECM molecule and, when bound, transmit a signal into the cell. Use the proteins shown in (c) and (f) to explain how this might occur.*

*For suggested answer, see*Appendix A.**

Animation

Functions of the Plasma Membrane

image

Play

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Proteins on a cell's surface are important in the medical field. For example, a protein called CD 4 on the surface of immune cells helps the human immunodeficiency virus (HIV) infect these cells, leading to acquired immune deficiency syndrome (AIDS). Despite multiple exposures to HIV, however, a small number of people do not develop AIDS and show no evidence of HIV-infected cells. Comparing their genes with the genes of infected individuals, researchers learned that resistant people have an unusual form of a gene that codes for an immune cell-surface protein called CCR5. Further work showed that although CD 4 is the main HIV receptor, HIV must also bind to CCR5 as a "co-receptor" to infect most cells **(Figure 7.8a)**. An absence of CCR5 on the cells of resistant individuals, due to the gene alteration, prevents the virus from entering the cells **(Figure 7.8b)**.

Figure 7.8

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The genetic basis for HIV resistance.
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The genetic basis for H I V resistance.

Part a. H I V can infect a cell with C C R 5 on its surface, as in most people. Image: H I V binds to a cell that has both C D 4 and C C R 5 surface protein receptors.

Part b. H I V cannot infect a cell lacking C C R 5 on its surface, as in resistant individuals. Image: H I V cannot bind to a cell that has C D 4 on its membrane but lacks C C R 5.

MAKE CONNECTIONS *Study**Figures 2.16**and*5.17*; each shows pairs of molecules binding to each other. What would you predict about CCR5 that would allow HIV to bind to it? How could a drug molecule interfere with this binding?*

*For suggested answer, see*Appendix A.**

This information has been key to developing a treatment for HIV infection. Interfering with CD 4 causes dangerous side effects because of its many important functions in cells. Discovery of the CCR5 co-receptor provided a safer target for development of drugs that mask this protein and block HIV entry. One such drug, maraviroc (brand name Selzentry), was approved for treatment of HIV in 2007; ongoing trials to determine its ability to prevent HIV infection in uninfected, at-risk patients have been disappointing.

The Role of Membrane Carbohydrates in Cell-Cell Recognition

Cell-cell recognition, a cell's ability to distinguish one type of neighboring cell from another, is crucial to the functioning of an organism. It is important, for example, in the sorting of cells into tissues and organs in an animal embryo. It is also the basis for the rejection of foreign cells by the immune system, an important line of defense in vertebrate animals (see Concept 43.1). Cells recognize other cells by binding to molecules, often containing carbohydrates, on the extracellular surface of the plasma membrane (see Figure 7.7d).

Membrane carbohydrates are usually short, branched chains of fewer than 15 sugar units. Some are covalently bonded to lipids, forming molecules called **glycolipids**. (Recall that *glyco* refers to carbohydrate.) However, most are covalently bonded to proteins, which are thereby ****__glycoproteins__ (see Figure 7.3).

The carbohydrates on the extracellular side of the plasma membrane vary from species to species, among individuals of the same species, and even from one cell type to another in a single individual. The diversity of the molecules and their location on the cell's surface enable membrane carbohydrates to function as markers that distinguish one cell from another. For example, the four human blood types designated A, B, AB, and O reflect variation in the carbohydrate part of glycoproteins on the surface of red blood cells.

Synthesis and Sidedness of Membranes

Membranes have distinct inside and outside faces. The two lipid layers may differ in lipid composition, and each protein has directional orientation in the membrane (see Figure 7.6). **Figure 7.9** shows how membrane sidedness arises: The asymmetrical arrangement of proteins, lipids, and their associated carbohydrates in the plasma membrane is determined as the membrane is being built.

Figure 7.9

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    Synthesis of membrane components and their orientation in
the membrane.
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Synthesis of membrane components in a series of four steps.

The cytoplasmic (orange) face of the plasma membrane differs from the extracellular (aqua) face. The latter arises from the inside face of ER, Golgi, and vesicle membranes.

Step 1: Secretory proteins, membrane proteins, and lipids are synthesized in the endoplasmic reticulum (E R). In the E R, carbohydrates (green blocks) are added to the transmembrane proteins (purple dumbbells across the E R lipid bilayer), making them glycoproteins. The carbohydrate portions may then be modified. Materials are transported in vesicles to the Golgi apparatus.

Step 2: Inside the Golgi apparatus, the glycoproteins undergo further carbohydrate modification, and lipids acquire carbohydrates, becoming glycolipids.

Step 3: The glycoproteins, glycolipids, and secretory proteins (purple spheres) are transported in vesicles to the plasma membrane.

Step 4: As vesicles fuse with the plasma membrane, the outside face of the vesicle becomes continuous with the inside (cytoplasmic) face of the plasma membrane. This releases the secretory proteins from the cell, a process called exocytosis, and positions the carbohydrates of membrane glycoproteins and glycolipids on the outside (extracellular) face of the plasma membrane.

DRAW IT *Draw an integral membrane protein extending from partway through the ER membrane into the ER lumen. Next, draw the protein where it would be located in a series of numbered steps ending at the plasma membrane. Would the protein contact the cytoplasm or the extracellular fluid? Explain.*

*For suggested answer, see*Appendix A.**

Concept 7.2: Membrane structure results in selective permeability

The biological membrane has emergent properties beyond those of the many individual molecules that make it up. The remainder of this chapter focuses on one of those properties: A membrane exhibits **selective permeability**; that is, it allows some substances to cross more easily than others. The ability to regulate transport across cellular boundaries is essential to the cell's existence. We will see once again that form fits function: The fluid mosaic model helps explain how membranes regulate the cell's molecular traffic.

A steady traffic of small molecules and ions moves across the plasma membrane in both directions. Consider the chemical exchanges between a muscle cell and the extracellular fluid that bathes it. Sugars, amino acids, and other nutrients enter the cell, and metabolic waste products leave it. The cell takes in O_2 for use in cellular respiration and expels CO_2. Also, the cell regulates its concentrations of inorganic ions, such as Na^+, K^+, Ca^{2+}, and Cl^-, by shuttling them one way or the other across the plasma membrane. In spite of heavy traffic through them, cell membranes are selective in their permeability: Substances do not cross the barrier indiscriminately. The cell is able to take up some small molecules and ions and exclude others.

The Permeability of the Lipid Bilayer

Nonpolar molecules, such as hydrocarbons, CO_2, and O_2, are hydrophobic, as are lipids. They can all therefore dissolve in the lipid bilayer of the membrane and cross it easily, without the aid of membrane proteins. However, the hydrophobic interior of the membrane impedes direct passage through the membrane of ions and polar molecules, which are hydrophilic. Polar molecules such as glucose and other sugars pass only slowly through a lipid bilayer, and even water, a very small polar molecule, does not cross rapidly relative to nonpolar molecules. A charged atom or molecule and its surrounding shell of water (see Figure 3.8) are even less likely to penetrate the hydrophobic interior of the membrane. Furthermore, the lipid bilayer is only one aspect of the gatekeeper system responsible for a cell's selective permeability. Proteins built into the membrane play key roles in regulating transport.

Transport Proteins

Specific ions and a variety of polar molecules can't move through cell membranes on their own. However, these hydrophilic substances can avoid contact with the lipid bilayer by passing through **transport proteins** that span the membrane.

Some transport proteins, called *channel proteins*, function by having a hydrophilic channel that certain molecules or ions use as a tunnel through the membrane (see Figure 7.7a, left). For example, the passage of water molecules through the membrane in certain cells is greatly facilitated by channel proteins known as **aquaporins(Figure 7.10)**. Most aquaporin proteins consist of four identical polypeptide subunits. Each polypeptide forms a channel that water molecules pass through, single-file, overall allowing entry of up to 3 billion water molecules per second. Without aquaporins, only a tiny fraction of these water molecules would pass through the same area of the cell membrane in a second. Other transport proteins, called *carrier proteins*, hold on to their passengers and change shape in a way that shuttles them across the membrane (see Figure 7.7a, right).

Figure 7.10

```
An aquaporin.
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Red and gray molecules pass through a blue ribbon in a lipid bilayer.

This computer model shows water molecules (red and gray) passing through an aquaporin (blue ribbons), in a lipid bilayer (yellow, hydrophilic heads; green, hydrophobic tails).

Interview

Interview with Peter Agre: Discovering aquaporins

A photo of Peter Agre.

Animation

Aquaporin

image

Play

image

Play

**

A transport protein is specific for the substance it translocates (moves), allowing only a certain substance (or a small group of related substances) to cross the membrane. For example, a glucose carrier protein in the plasma membrane of red blood cells transports glucose across the membrane 50,000 times faster than glucose can pass through on its own. This “glucose transporter” is so selective that it even rejects fructose, a structural isomer of glucose (see Figure 5.3). Thus, the selective permeability of a membrane depends on both the discriminating barrier of the lipid bilayer and the specific transport proteins built into the membrane.

Animation

Selective Permeability of Membranes

image

Play

image

Play

**

What establishes the *direction* of traffic across a membrane? And what mechanisms drive molecules across membranes? We will address these questions next as we explore two modes of membrane traffic: passive transport and active transport.

Concept 7.3: Passive transport is diffusion of a substance across a membrane with no energy investment

Molecules have a type of energy called thermal energy, due to their constant motion (see Concept 3.2). One result of this motion is **diffusion**, the movement of particles of any substance so that they spread out into the available space. Each molecule moves randomly, yet diffusion of a *population* of molecules may be directional. Imagine a synthetic membrane separating pure water from a solution of a dye in water. Study **Figure 7.11a** carefully to see how diffusion would result in equal concentrations of dye molecules in both solutions. At that point, a dynamic equilibrium will exist, with as many dye molecules crossing per second in one direction as in the other.

Figure 7.11

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Diffusion of solutes across a synthetic membrane.
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Diffusion of solutes across a synthetic membrane.

Each large arrow shows net diffusion of dye molecules of that color.

Parts are as follows.

Part a. Diffusion of one solute. The membrane has pores large enough for molecules of dye to pass through. Random movement of dye molecules will cause some to pass through the pores; this will happen more often on the side with more dye molecules. The dye diffuses from where it is more concentrated to where it is less concentrated (called diffusing down a concentration gradient). This leads to a dynamic equilibrium: The solute molecules continue to cross the membrane, but at roughly equal rates in both directions. In the first image, there is a high concentration of dye molecules on one side of a synthetic membrane and no dye molecules on the other side. Small arrows within the image indicate that the dye molecules are in motion, and an arrow beneath the image indicates that the net direction for the diffusion of the dye molecules is from high concentration to low (no) concentration across the membrane. In the second image, some dye molecules have diffused across the membrane, but the molecules are still more concentrated on one side. Arrows indicate that the direction of net diffusion will continue as dye molecules move from high to low concentration across the membrane. In the final image the concentration of dye molecules is equal on both

sides of the membrane. Arrows indicate that the dye molecules are now at equilibrium, meaning they will still show motion, but there will be no more net movement of dye molecules across the membrane.

Part b. Diffusion of two solutes. Solutions of two different dyes are separated by a membrane that is permeable to both. Each dye diffuses down its own concentration gradient. There will be a net diffusion of the purple dye toward the left, even though the total solute concentration was initially greater on the left side. In the first image, one molecule is concentrated on one side of a membrane, while a different molecule is concentrated on the other side. Arrows indicate that both molecules are in motion, and that each will begin to diffuse down its own concentration gradient. In the second image, diffusion progresses independently for each molecule. Both molecules diffuse across the membrane down their own concentration gradients regardless of the concentration of the other molecule. In the third image, equilibrium for both molecules has been reached, with both equally concentrated on both sides of the membrane.

Animation

Diffusion

image

image

**

Here is a simple rule of diffusion: In the absence of any other forces, a substance will diffuse from where it is more concentrated to where it is less concentrated. Put another way, a substance diffuses down its **concentration gradient**, the region along which the density of a chemical substance increases or decreases (in this case, decreases). Diffusion is a spontaneous process, needing no input of energy. Each substance diffuses down its *own* concentration gradient, unaffected by the concentration gradients of other substances **(Figure 7.11b)**.

Much of the traffic across cell membranes occurs by diffusion. When a substance is more concentrated on one side of a membrane than on the other, there is a tendency for it to diffuse across, down its concentration gradient (assuming that the membrane is permeable to that substance). One important example is the uptake of oxygen by a cell performing cellular respiration. Dissolved oxygen diffuses into the cell across the plasma membrane. As long as cellular respiration consumes the O2 as it enters, diffusion into the cell will continue because the concentration gradient favors movement in that direction.

The diffusion of a substance across a biological membrane is called **passive transport** because it requires no energy. The concentration gradient itself represents potential energy (see Concept 2.2 and Figure 8.5b) and drives diffusion. Remember, though, that membranes are selectively permeable and therefore have

different effects on the rates of diffusion of various molecules. Water can diffuse very rapidly across the membranes of cells with aquaporins, compared with diffusion in the absence of aquaporins. The movement of water across the plasma membrane has important consequences for cells.

Effects of Osmosis on Water Balance

To see how two solutions with different solute concentrations interact, picture a U-shaped glass tube with a selectively permeable artificial membrane separating two sugar solutions **(Figure 7.12)**. Pores in this synthetic membrane are too small for sugar molecules to pass through but large enough for water molecules. However, tight clustering of water molecules around the hydrophilic solute molecules makes some of the water unavailable to cross the membrane. As a result, the solution with a higher solute concentration has a lower *free* water concentration. Water diffuses across the membrane from the region of higher free water concentration (lower solute concentration) to that of lower free water concentration (higher solute concentration) until the solute concentrations on both sides of the membrane are more nearly equal. The diffusion of free water across a selectively permeable membrane, whether artificial or cellular, is called **osmosis**. The movement of water across cell membranes and the balance of water between the cell and its environment are crucial to organisms. Let's now apply what we've learned about osmosis in this system to living cells.

Figure 7.12

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Osmosis.
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An illustration of osmosis.

Two sugar solutions of different concentrations are separated by a membrane that the solvent (water) can pass through but the solute (sugar) cannot. Water molecules move randomly and may cross in either direction, but overall, water diffuses from the solution with less concentrated solute to that with more concentrated solute. This passive transport of water, or osmosis, makes the sugar concentrations on both sides roughly equal.

Two U shaped tubes filled with solutions of different sugar concentrations separated by a selectively permeable membrane at the base of the U tube are depicted. The first U shaped tube shows the initial starting point of the experiment where fluid levels on either side of the membrane are equal. Arrows indicate that water will move by osmosis across the membrane from the solution with lower solute (sugar) concentration to the solution with higher solute concentration. The result, shown in a second image depicting the U shaped tube, is that as water moves across the membrane from low to high solute

concentration, the fluid level rises with the influx of water, and is higher than the fluid level on the other side of the membrane. After osmosis, although the fluid level is higher on the side of the membrane that originally had the higher solute concentration, solute concentrations are similar on both sides of the membrane. A third image is depicted by zooming in on the membrane area of the U shaped tube and gives a closer look at water and solute molecules during osmosis on both sides of the membrane. On the side of the membrane with a lower solute concentration, there are fewer solute molecules and more free water molecules (water molecules not interacting with solute molecules). From this side, water molecules can pass through pores, but sugar molecules cannot. On the side of the membrane with a higher solute concentration, there are more solute molecules and fewer free water molecules, as more water molecules are clustered around the sugar molecules rather than moving freely. Thus, osmosis proceeds from lower solute concentration to higher solute concentration, as there are more free water molecules in the solution with lower solute concentration that can move across the membrane. Water moves from an area of higher to lower free water concentration (lower to higher solute concentration).

VISUAL SKILLS *If an orange dye capable of passing through the membrane was added to the left side of the tube above, how would it be distributed at the end of the experiment? (See*Figure 7.11*.)* Would the final solution levels in the tube be affected? What cellular component does the membrane represent in this experiment?*

*For suggested answer, see*Appendix A*.*

Figure 7.12 Walkthrough

Osmosis

Watch an explanation of this figure in the Figure 7.12 Walkthrough: Osmosis.

image

Play

image

Water Balance of Cells Without Cell Walls

To explain the behavior of a cell in a solution, we must consider both solute concentration and membrane permeability. Both factors are taken into account in the concept of **tonicity**, the ability of a surrounding solution to cause a cell to gain or lose water. The tonicity of a solution depends in part on its concentration of solutes that cannot cross the membrane (nonpenetrating solutes) relative to that inside the cell. If there is a higher concentration of nonpenetrating solutes in the surrounding solution, water will tend to leave the cell, and vice versa.

If a cell without a cell wall, such as an animal cell, is immersed in an environment that is **isotonic** to the cell (*iso* means “same”), there will be no *net* movement of water across the plasma membrane. Water diffuses across the membrane, but at the same rate in both directions. In an isotonic environment, the volume of an animal cell is stable **(Figure 7.13a)**.

Figure 7.13

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The water balance of living cells.
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The water balance of living animal and plant cells.

How living cells react to changes in the solute concentration of their environment depends on whether or not they have cell walls. ***(a)*** *Animal cells, such as this red blood cell, do not have cell walls.* ***(b)*** *Plant cells do have cell walls. (Arrows indicate net water movement after the cells were first placed in these solutions.)*

Part a. Animal cell. An animal cell fares best in an isotonic environment unless it has special adaptations that offset the osmotic uptake or loss of water. Water flows into animal cells in hypotonic solutions, and may result is cells lysing (bursting). There is no net flow of water in or out of the cell in isotonic solutions; this condition is normal for animal cells. In hypertonic solutions, water leaves the animal cell and the animal cell shrivels.

Part b. Plant cell. Plant cells are turgid (firm) and generally healthiest in a hypotonic environment, where the uptake of water is eventually balanced by the wall pushing back on the cell. Water flows into plant cells in hypotonic solutions, creating turgid (normal) conditions for plant cells as water pushes against the cell wall. In isotonic solutions, there is no net flow of water in or out of the cell and the plant cell is flaccid. In hypertonic solutions, water flows out of the cell and the plant cell can be plasmolyzed.

Why do limp celery stalks become crisp when placed in a glass of water?

*For suggested answer, see*Appendix A.**

Animation

Osmosis and Water Balance in Cells

image

Play

image

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Video

Turgid*Elodea*

image

image

Video

Plasmolysis in *Elodea*

image

image

Let's transfer the cell to a solution that is **hypertonic** to the cell (*hyper* means "more," in this case referring to nonpenetrating solutes). The cell will lose water, shrivel, and probably die. This is why an increase in the salinity (saltiness) of a lake can kill the animals there; if the lake water becomes hypertonic to the animals' cells, they might shrivel and die. However, taking up too much water can be just as hazardous to a cell as losing water. If we place the cell in a solution that is **hypotonic** to the cell (*hypo* means "less"), water will enter the cell faster than it leaves, and the cell will swell and lyse (burst) like an overfilled water balloon.

A cell without rigid cell walls can't tolerate either excessive uptake or excessive loss of water. This problem of water balance is automatically solved if such a cell lives in isotonic surroundings. Seawater is isotonic to many marine invertebrates. The cells of most terrestrial (land-dwelling) animals are bathed in an extracellular fluid that is isotonic to the cells. In hypertonic or hypotonic environments, however, organisms that lack rigid cell walls must have other adaptations for **osmoregulation**, the control of solute concentrations and water balance. For example, the unicellular protist *Paramecium caudatum* lives in pond water, which is hypotonic to the cell. *Paramecium* has a plasma membrane that is much less permeable to water than the membranes of most other cells, but this only slows the uptake of water, which continually enters the cell. The reason the *Paramecium* cell doesn't burst is that it has a contractile vacuole, an organelle that functions as a pump to force water out of the cell as fast as it enters by osmosis **(Figure 7.14)**. In contrast, the bacteria and archaea that live in hypersaline (excessively salty) environments (see Figure 27.1) have cellular mechanisms that balance the internal and external solute concentrations to ensure that water does not move out of the cell. We'll examine other evolutionary adaptations for osmoregulation by animals in Concept 44.1.

Figure 7.14

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The contractile vacuole of *Paramecium*.
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A Paramecium with contractile vacuoles is shown on a 50 micrometer scale. It is approximately 200 micrometers in length.

The vacuole collects fluid from canals in the cytoplasm. When full, the vacuole and canals contract, expelling fluid from the cell (LM).

Video

Paramecium Vacuole

image

image

Water Balance of Cells with Cell Walls

The cells of plants, prokaryotes, fungi, and some protists are surrounded by cell walls (see Figure 6.27). When such a cell is immersed in a hypotonic solution—bathed in rainwater, for example—the cell wall helps maintain the cell's water balance. Consider a plant cell. Like an animal cell, the plant cell swells as water enters by osmosis **(Figure 7.13b)**. However, the relatively inelastic cell wall will expand only so much before it exerts a back pressure on the cell, called *turgor pressure*, that opposes further water uptake. At this point, the cell is **turgid** (very firm), which is the healthy state for most plant cells. Plants that are not woody, such as most houseplants, depend for mechanical support on cells kept turgid by a surrounding hypotonic solution. If a plant's cells and surroundings are isotonic, there is no net tendency for water to enter and the cells become **flaccid** (limp); the plant wilts.

However, a cell wall is of no advantage if the cell is immersed in a hypertonic environment. In this case, a plant cell, like an animal cell, will lose water to its surroundings and shrink. As the plant cell shrivels, its plasma membrane pulls away from the cell wall at multiple places. This phenomenon, called **plasmolysis**, causes the plant to wilt and can lead to plant death. The walled cells of bacteria and fungi also plasmolyze in hypertonic environments.

Facilitated Diffusion: Passive Transport Aided by Proteins

Let's look more closely at how water and certain hydrophilic solutes cross a membrane. As mentioned earlier, many polar molecules and ions blocked by the lipid bilayer of the membrane diffuse passively with the help of transport proteins that span the membrane. This phenomenon is called **facilitated diffusion**.

Cell biologists are still trying to learn exactly how various transport proteins facilitate diffusion. Most transport proteins are very specific: They transport some substances but not others.

As mentioned earlier, the two types of transport proteins are channel proteins and carrier proteins. Channel proteins simply provide corridors that allow specific molecules or ions to cross the membrane **(Figure 7.15a)**. The hydrophilic passageways provided by these proteins can allow water molecules or small ions to diffuse very quickly from one side of the membrane to the other. Aquaporins, the water channel proteins, facilitate the massive levels of diffusion of water (osmosis) that occur in plant cells and in animal cells such as red blood cells (see Figure 7.13). Certain kidney cells also have a high number of aquaporins, allowing them to reclaim water from urine before it is excreted. If the kidneys didn't perform this function, you would excrete about 180 L of urine per day—and have to drink an equal volume of water!

Figure 7.15

```
    Two types of transport proteins that carry out facilitated
diffusion.
```

Diagrams of a channel protein and carrier protein.

In both cases, the protein can transport the solute in either direction, but the net movement is down the concentration gradient of the solute.

Part a. A channel protein has a channel through which water molecules or a specific solute can pass. Water molecules move across the membrane from higher concentration in the extracellular fluid to lower concentration in the cytoplasm through a hydrophilic channel in a channel protein.

Part b. A carrier protein alternates between two shapes, moving a solute across the membrane during the shape change. Solutes move from higher concentration in the extracellular fluid to lower concentration in the cytoplasm by moving through a carrier protein in the membrane, which changes shape to transport them.

Animation

Facilitated Diffusion

image

image

**

Channel proteins that transport ions are called **ion channels**. Many ion channels function as **gated channels**, which open or close in response to a stimulus (see Figure 11.8). For some gated channels, the stimulus is electrical. In a nerve cell, for example, a potassium ion channel protein (see computer model) opens in response to an electrical stimulus, allowing a stream of potassium ions to leave the cell. This restores the cell's ability to fire again. Other gated channels have a chemical stimulus: They open or close when a specific substance (not the one to be transported) binds to the channel. Ion channels are important in the functioning of the nervous system, as you'll learn in Chapter 48.

A potassium ion is at the center of a protein.

Potassium ion channel protein (See also the side view of a calcium channel protein in Figure 6.32a.)

Interview

Interview with Elba Serrano: Investigating how ion channels enable you to hear

A photo of Elba Serrano.

Carrier proteins, such as the glucose transporter mentioned earlier, seem to undergo a subtle change in shape that somehow translocates the solute-binding site across the membrane **(Figure 7.15b)**. Such a change in shape may be triggered by the binding and release of the transported molecule. Like ion channels, carrier proteins involved in facilitated diffusion result in the net movement of a substance down its concentration gradient. No energy input is thus required: This is passive transport. The **Scientific Skills Exercise** gives you an opportunity to work with data from an experiment related to glucose transport.

BioFlix® Animation

Passive Transport

image

Scientific Skills Exercise

Interpreting a Scatter Plot with Two Sets of Data

AP Science Practices 2, 3, 4, 6

Is Glucose Uptake into Cells Affected by Age? Glucose, an important energy source for animals, is transported into cells by facilitated diffusion using protein carriers. In this exercise, you will interpret a graph with two sets of data from an experiment that examined glucose uptake over time in red blood cells from guinea pigs of different ages. You will determine if the cells' rate of glucose uptake depended on the age of the guinea pig.

How the Experiment Was Done Researchers incubated guinea pig red blood cells in a 300 mM (millimolar) radioactive glucose solution at pH 7.4 at 25°C. Every 10 or 15 minutes, they removed a sample of cells and measured the concentration of radioactive glucose inside those cells. The cells came from either a 15-day-old or a 1-month-old guinea pig.

Data from the Experiment When you have multiple sets of data, it can be useful to plot them on the same graph for comparison. In the graph here, each set of dots (of the same color) forms a *scatter plot*, in which every data point represents two numerical values, one for each variable. For each data set, a curve that best fits the points has been drawn to make it easier to see the trends. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)

Two graphs plot concentration of radioactive glucose versus Incubation time in minutes.

Glucose Uptake over Time in Guinea Pig Red Blood Cells.

Data from T. Kondo and E. Beutler, Developmental changes in glucose transport of guinea pig erythrocytes, *Journal of Clinical Investigation* 65:1–4 (1980).

The graph of a 15-day-old guinea pig rises from (0, 0) through (10, 50), (20, 72), (30, 85), and (50, 97) to (60, 102). The graph for a 1-month-old-guinea pig rises from (0, 0) through (10, 20), (20, 35), (30, 40), and (45, 48) to (60, 55). All values are estimated.

INTERPRET THE DATA

- First make sure you understand the parts of the graph.
- Which variable is the independent variable—the variable controlled by the researchers?
- Which variable is the dependent variable—the variable that depended on the treatment and was measured by the researchers?
- What do the red dots represent?
- The blue dots?
- From the data points on the graph, construct a table of the data. Put "Incubation Time (min)" in the left column of the table.

- What does the graph show? Compare and contrast glucose uptake in red blood cells from 15-day-old and 1-month-old guinea pigs.
- Develop a hypothesis to explain the difference between glucose uptake in red blood cells from 15-day-old and 1-month-old guinea pigs. (Think about how glucose gets into cells.)
- Design an experiment to test your hypothesis.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Concept 7.4: Active transport uses energy to move solutes against their gradients

Despite the help of transport proteins, facilitated diffusion is considered passive transport because the solute is moving down its concentration gradient, a process that requires no energy. Facilitated diffusion speeds transport of a solute by providing efficient passage through the membrane, but it does not alter the direction of transport. Some other transport proteins, however, can use energy to move solutes *against* their concentration gradients, across the plasma membrane from the side where they are less concentrated (whether inside or outside) to the side where they are more concentrated.

The Need for Energy in Active Transport

To pump a solute across a membrane against its gradient requires work; the cell must expend energy. Therefore, this type of membrane traffic is called **active transport**. The transport proteins that move solutes against their concentration gradients are all carrier proteins rather than channel proteins. This makes sense because when channel proteins are open, they merely allow solutes to diffuse down their concentration gradients rather than picking them up and transporting them against their gradients.

Active transport enables a cell to maintain internal concentrations of small solutes that differ from concentrations in its environment. For example, compared with its surroundings, an animal cell has a much higher concentration of potassium ions (K+) and a much lower concentration of sodium ions (Na+). The plasma membrane helps maintain these steep gradients by pumping Na+ out of the cell and K+ into the cell.

As in other types of cellular work, ATP hydrolysis supplies the energy for most active transport. One way ATP can power active transport is when its terminal phosphate group is transferred directly to the transport

protein. This can induce the protein to change its shape in a manner that translocates a solute bound to the protein across the membrane. One transport system that works this way is the **sodium-potassium pump**, which exchanges Na+ for K+ across the plasma membrane of animal cells **(Figure 7.16)**. The distinction between passive transport and active transport is reviewed in **Figure 7.17**.

Figure 7.16

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    The sodium–potassium pump: a specific case of active
transport.
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An illustration depicting the working of the sodium-potassium pump.

The sodium potassium pump is a specialized carrier protein in the cell membrane that performs active transport of sodium and potassium ions. Steps are as follows.

- In the extracellular fluid, sodium concentration is high and potassium concentration is low compared with the cytoplasm on the other side of the membrane. Three cytoplasmic sodium ions bind with the carrier protein, as in this configuration the protein has a high affinity for sodium ions.
- Binding of the three sodium ions to the carrier protein stimulates phosphorylation by A T P using a kinase enzyme. A phosphate group is transferred from A T P to the carrier protein, leaving A D P in the cytoplasm.
- Phosphorylation leads to a change in protein shape, reducing its affinity for sodium ions, which are released outside the cell.
- The protein's new shape has a high affinity for potassium ions, which bind to the protein on the extracellular side (2 potassium ions bind). This triggers the release of the phosphate group inside the cell.
- Loss of the phosphate group restores the protein's original shape, which has a lower affinity for potassium ions.
- The two potassium ions are released inside the cell; the protein's affinity for sodium ions is high again, and the cycle repeats.

This transport system pumps ions against steep concentration gradients. The pump oscillates between two shapes in a cycle that moves Na+ out of the cell (**STEPS 1–3**) and K+ into the cell (**STEPS 4–6**). The two shapes have different binding affinities for Na+ and K+ ATP hydrolysis powers the shape change by transferring a phosphate group to the transport protein (phosphorylating the protein).

VISUAL SKILLS *For each ion (Na+andK+), describe its concentration inside the cell relative to outside. How many*Na+*are moved out of the cell and how many*K+*moved in per cycle?*

*For suggested answer, see*Appendix A.**

Animation

Active Transport

image

Play

image

Play

**

Figure 7.17

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    Review: passive and active transport.
```

Passive versus active transport.

Passive transport. Substances diffuse spontaneously down their concentration gradients, crossing a membrane with no expenditure of energy by the cell. The rate of diffusion can be greatly increased by transport proteins in the membrane. Mechanisms of passive transport include diffusion and facilitated diffusion.

- Diffusion. Hydrophobic molecules and (at a slow rate) very small uncharged polar molecules can diffuse through the lipid bilayer.
- Facilitated diffusion. Many hydrophilic substances diffuse through membranes with the assistance of transport proteins, either channel proteins (left) or carrier proteins (right).

Active transport. Some transport proteins act as pumps, moving substances across a membrane against their concentration (or electrochemical) gradients. Energy is usually supplied by A T P hydrolysis.

VISUAL SKILLS *For each solute in the right panel, describe its direction of movement, and state whether it is moving with or against its concentration gradient.*

*For suggested answer, see*Appendix A.**

How Ion Pumps Maintain Membrane Potential

All cells have voltages across their plasma membranes. Voltage is electrical potential energy (see Concept 2.2)—a separation of opposite charges. The cytoplasmic side of the membrane is negative in charge relative to the extracellular side because of an unequal distribution of anions and cations on the two sides. The voltage across a membrane, called a **membrane potential**, ranges from about −50 to −200 millivolts (mV). (The minus sign indicates that the inside of the cell is negative relative to the outside.)

The membrane potential acts like a battery, an energy source that affects the traffic of all charged substances across the membrane. Because the inside of the cell is negative compared with the outside, the membrane potential favors the passive transport of cations into the cell and anions out of the cell. Thus, *two* forces drive the diffusion of ions across a membrane: a chemical force (the ion's concentration gradient, which has been our sole consideration thus far in the chapter) and an electrical force (the effect of the membrane potential on the ion's movement). This combination of forces acting on an ion is called the **electrochemical gradient**.

In the case of ions, then, we must refine our concept of passive transport: An ion diffuses not simply down its *concentration* gradient but, more exactly, down its *electrochemical* gradient. For example, the concentration of Na+ inside a resting nerve cell is much lower than outside it. When the cell is stimulated, gated channels open that facilitate Na+ diffusion. Sodium ions then "fall" down their electrochemical gradient, driven by the concentration gradient of Na+ and by the attraction of these cations to the negative side (inside) of the membrane. In this example, both electrical and chemical contributions to the electrochemical gradient act in the same direction across the membrane, but this is not always so. In cases where electrical forces due to the membrane potential oppose the simple diffusion of an ion down its concentration gradient, active transport may be necessary. In Concepts 48.2 and 48.3, you'll learn about the importance of electrochemical gradients and membrane potentials in the transmission of nerve impulses.

Some membrane proteins that actively transport ions contribute to the membrane potential. Study Figure 7.16 to see if you can see why the sodium-potassium pump is a good example. Notice that the pump does not translocate Na+ and K+ one for one, but pumps *three* sodium ions out of the cell for every *two* potassium ions it pumps into the cell. With each "crank" of the pump, there is a net transfer of one positive charge from the cytoplasm to the extracellular fluid, a process that stores energy as voltage. A transport protein that generates voltage across a membrane is called an **electrogenic pump**. The sodium-potassium pump appears to be the major electrogenic pump of animal cells. The main electrogenic pump of plants, fungi, and bacteria is a **proton pump**, which actively transports protons (hydrogen ions, H+) out of the cell. The pumping of H+ transfers positive charge from the cytoplasm to the extracellular solution **(Figure 7.18)**. By generating voltage across membranes, electrogenic pumps help store energy that can be tapped for cellular

work. One important use of proton gradients in the cell is for ATP synthesis during cellular respiration, as you will see in Concept 9.4. Another is a type of membrane traffic called cotransport.

Figure 7.18

A proton pump.

A T P is used to actively transport protons out of the cell through a proton pump. Negative charges shown in the cytoplasm and positive charges in the extracellular fluid.

Proton pumps are electrogenic pumps that store energy by generating voltage (charge separation) across membranes. A proton pump translocates positive charge in the form of hydrogen ions (H+or protons). The voltage and H+concentration gradient represent a dual energy source that can drive other processes, such as the uptake of nutrients. Most proton pumps are powered by ATP hydrolysis. (See Figure 6.32a.)

Cotransport: Coupled Transport by a Membrane Protein

A solute that exists in different concentrations across a membrane can do work as it moves across that membrane by diffusion down its concentration gradient. This is analogous to water that has been pumped uphill and performs work as it flows back down. In a mechanism called **cotransport**, a transport protein (a cotransporter) can couple the “downhill” diffusion of the solute to the “uphill” transport of a second substance against its own concentration gradient. For instance, a plant cell uses the gradient of H+ generated by its ATP-powered proton pumps to drive the active transport of amino acids, sugars, and several other nutrients into the cell. In the example shown in **Figure 7.19**, a cotransporter couples the return of H+ to the transport of sucrose into the cell. This protein can translocate sucrose into the cell against its concentration gradient, but only if the sucrose molecule travels in the company of an H+. The H+ uses the transport protein as an avenue to diffuse down its own electrochemical gradient, which is maintained by the proton pump. Plants use H+/sucrose cotransport to load sucrose produced by photosynthesis into cells in the veins of leaves. The vascular tissue of the plant can then distribute the sugar to roots and other nonphotosynthetic organs that do not make their own sugars.

Figure 7.19

Cotransport: active transport driven by a concentration gradient.

A diagram of co-transport.

Diagram shows cell membrane with Sucrose-H plus cotransporter and proton pump imbedded. H plus ions are more concentrated in the extracellular fluid than inside the cell. Arrows indicate that protons are pumped across the membrane out of the cell through the proton pump using A T P. Arrows indicate that

protons can diffuse down their concentration gradient back into the cell through the Sucrose-H plus cotransporter, and at the same time sucrose (a disaccharide sugar composed of two ring structures) is transported into the cell through the Sucrose-H plus co-transporter against its concentration gradient.

A carrier protein, such as this H+/sucrose cotransporter in a plant cell (top), is able to use the diffusion of H+down its electrochemical gradient into the cell to drive the uptake of sucrose. (The cell wall is not shown.) Although not technically part of the cotransport process, an ATP-driven proton pump is shown here (bottom), which concentrates H+outside the cell. The resulting H+gradient represents potential energy that can be used for active transport—of sucrose, in this case. Thus, ATP hydrolysis indirectly provides the energy necessary for cotransport.

A similar cotransporter in animals transports glucose into intestinal cells together with Na+, which is moving down its concentration gradient into the cell. The glucose and Na+ then exit the cell into the blood on the opposite side, the glucose by facilitated diffusion and the Na+ by active transport (see Figure 7.16). Water follows these solutes into the cell, and then into the blood, via osmosis. Our understanding of Na+/glucose cotransporters has helped us to find effective treatments for severe diarrhea, a serious problem that can cause life-threatening dehydration even if patients drink large volumes of water. To treat diarrhea, patients are given a solution to drink containing therapeutic concentrations of salt (NaCl) and glucose. The solutes are taken up by Na+/glucose cotransporters and passed through the intestinal cells into the blood, followed by water via osmosis. Without the solutes, sufficient water cannot be absorbed from the intestine. In developing countries, where diarrheal illnesses in infants and young children are a substantial problem, this simple treatment has reduced mortality.

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Active Transport

image

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image

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Concept 7.5: Bulk transport across the plasma membrane occurs by exocytosis and

endocytosis

Large molecules, such as proteins and polysaccharides, generally don't cross the membrane by diffusion or transport proteins. Instead, they usually enter and leave the cell in bulk, packaged in vesicles.

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Exocytosis and Endocytosis

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Exocytosis

The cell secretes certain molecules by the fusion of vesicles with the plasma membrane; this process is called **exocytosis(Figure 7.20)**. A transport vesicle that has budded from the Golgi apparatus moves along a microtubule of the cytoskeleton to the plasma membrane. When the vesicle membrane and plasma membrane come into contact, specific proteins in both membranes rearrange the lipid molecules of the two bilayers so that the two membranes fuse. The contents of the vesicle spill out of the cell, and the vesicle membrane becomes part of the plasma membrane.

Figure 7.20

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Exocytosis.
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A diagram of the process of movement of vesicles, leading to exocytosis.

The process of exocytosis occurs as follows:

- A vesicle made of plasma membrane components, containing secretory proteins, buds off from the Golgi apparatus.
- The vesicle travels along a microtubule to the plasma membrane.
- The outside of the vesicle binds to the inside of the plasma membrane.
- The two membranes fuse, releasing (secreting) the proteins outside the cell.

- The vesicle membrane becomes part of the plasma membrane.

Many secretory cells use exocytosis to export products. For example, cells in the pancreas that make insulin secrete it into the extracellular fluid by exocytosis. In another example, nerve cells use exocytosis to release neurotransmitters that signal other neurons or muscle cells (see Figure 7.1). When plant cells are making cell walls, exocytosis delivers some of the necessary proteins and carbohydrates from Golgi vesicles to the outside of the cell.

Endocytosis

In **endocytosis**, the cell takes in molecules and particulate matter by forming new vesicles from the plasma membrane. Although the proteins involved in the processes are different, the events of endocytosis look like the reverse of exocytosis. First, a small area of the plasma membrane sinks inward to form a pocket. Then, as the pocket deepens, it pinches in, forming a vesicle containing material that had been outside the cell. Study **Figure 7.21** carefully to understand the three types of endocytosis: **phagocytosis** ("cellular eating"), **pinocytosis** ("cellular drinking"), and **receptor-mediated endocytosis**.

Figure 7.21

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Exploring Endocytosis in Animal Cells
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Exploring endocytosis in Animal Cells. Three forms of endocytosis are described: phagocytosis, pinocytosis, and receptor-mediated endocytosis.

In phagocytosis, a cell engulfs a particle by extending pseudopodia (singular, pseudopodium) around it and packaging it within a membranous sac called a food vacuole. The particle will be digested after the food vacuole fuses with a lysosome containing hydrolytic enzymes (see Figure 6.13). Image: An amoeba engulfing a green algal cell via phagocytosis (TEM). At about 9 μm long, the algal cell is almost half the length of the amoeba.

In pinocytosis, a cell continually "gulps" droplets of extracellular fluid into tiny vesicles, formed by infoldings of the plasma membrane. In this way, the cell obtains molecules dissolved in the droplets. Because any and all solutes are taken into the cell, pinocytosis as shown here is nonspecific for the substances it transports. In many cases, as above, the parts of the plasma membrane that form vesicles are lined on their cytoplasmic side by a fuzzy layer of coat protein; the "pits" and resulting vesicles are said to be "coated." Image: Pinocytotic vesicles form as coated infoldings of the plasma membrane (scale: 25 micrometers. TEM).

Receptor-mediated endocytosis is a specialized type of pinocytosis that enables the cell to acquire bulk quantities of specific substances, even though those substances may not be very concentrated in the extracellular fluid. Embedded in the plasma membrane are proteins with receptor sites exposed to the extracellular fluid. Specific solutes bind to the receptors. The receptor proteins then cluster in coated pits, and each coated pit forms a vesicle containing the bound molecules. The diagram shows only bound molecules (purple triangles) inside the vesicle, but other molecules from the extracellular fluid are also present. After the ingested material is liberated from the vesicle, the emptied receptors are recycled to the plasma membrane by the same vesicle (not shown). Two solutes are pictured outside the cell, shown as circles and triangles. Pits coated with protein receptor sites take in the “triangle” solutes in much greater quantities than the “circle” solutes. Each coated pit containing a mixture of predominantly triangle solutes then becomes a vesicle inside the cell. Plasma membrane with coat proteins as well as a coated vesicle forming are visible in T E M S (scale 0.25 micrometers; about the diameter of the coated vesicle).

VISUAL SKILLS *Use the scale bars to estimate the diameters of (a) the food vacuole that will form around the algal cell (left micrograph; measure the length, not the width) and (b) the coated vesicle (lower right micrograph). (c) Which is larger, and by what factor?*

*For suggested answer, see*Appendix A.**

Animation

Exocytosis and Endocytosis

image

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image

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Human cells use receptor-mediated endocytosis to take in cholesterol for membrane synthesis and the synthesis of other steroids. Cholesterol travels in the blood in particles called low-density lipoproteins (LDLs), each a complex of lipids and a protein. LDLs bind to LDL receptors on plasma membranes and then enter the cells by endocytosis. In the inherited disease familial hypercholesterolemia, characterized by a very high level of cholesterol in the blood, LDLs cannot enter cells because the LDL receptor proteins are defective or missing. Cholesterol thus accumulates in the blood, contributing to early atherosclerosis, the buildup of lipids within blood vessel walls. This narrows the space in the vessels and impedes blood flow, potentially resulting in heart damage and stroke.

Endocytosis and exocytosis also provide mechanisms for rejuvenating or remodeling the plasma membrane. These processes occur continually in most eukaryotic cells, yet the amount of plasma membrane in a nongrowing cell remains fairly constant. The addition of membrane by one process appears to offset the loss of membrane by the other.

BioFlix® Animation

Membrane Transport

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